MTH 204 : Lecture 17

Frobenius Method: Let $b(x)$ and $c(x)$ be any functions that are analytic at $x=0$ (i.e. they have no singularities at $x=0$)

Then the ODE $$y'' + \frac{b(x)}{x}y' + \frac{c(x)}{x^2}y = 0$$

has atleast one solution that can be represented in the form
$$y = x^r \sum_{m=0}^{\infty} a_m x^m \quad (a_0 \neq 0)$$

where r may be any (real or complex) numbers (and r is chosen so that $a_0 \neq 0$)
The ODE also has a second solution (such that these two solutions are linearly independent) that may be similar to the previous one (with different r and coefficients) or may contain a logarithmic term.

Ex: **Bessel's Equation:**

$$y'' + \frac{1}{x} y' + \frac{x^2 - \nu^2}{x^2} y = 0$$

(Here $b(x) = 1$, $c(x) = x^2 - \nu^2$)

Ex: Hypergeometric equation:

$$x(1-x)y'' + [c - (a+b+1)x]y' - aby = 0$$

(where a, b, c are constants)

Indicial Equation:

Consider the ODE: $y'' + \frac{b(x)}{x} y' + \frac{c(x)}{x^2} y = 0$

Multiplying by x^2 gives $x^2 y'' + x\, b(x) y' + c(x) y = 0$

If $b(x)$ and $c(x)$ are not polynomials, we expand $b(x)$ and $c(x)$ in power series

$$b(x) = b_0 + b_1 x + b_2 x^2 + \cdots\cdots$$

$$c(x) = c_0 + c_1 x + c_2 x^2 + \cdots\cdots$$

If $y = x^r \sum_{m=0}^{\infty} a_m x^m = x^r (a_0 + a_1 x + \cdots\cdots)$

is a solution, then

$$y' = \sum_{m=0}^{\infty} (m+r) a_m x^{m+r-1} = x^{r-1} \sum_{m=0}^{\infty} (m+r) a_m x^m$$

$$= x^{r-1} (r a_0 + (r+1) a_1 x \cdots\cdots)$$

$$y'' = \sum_{m=0}^{\infty} (m+r)(m+r-1) a_m x^{m+r-2}$$

$$= x^{r-2} \sum_{m=0}^{\infty} (m+r)(m+r-1) a_m x^m$$

$$= x^{r-2} [r(r-1) a_0 + (r+1) r a_1 x + \cdots]$$

Now substituting in $x^2 y'' + x b y' + c y = 0$

we obtain

$$\left. \begin{aligned} & x^r [r(r-1) a_0 + \cdots] + (b_0 + b_1 x + \cdots) x^r (r a_0 + \cdots) \\ & + (c_0 + c_1 x + \cdots\cdots) x^r (a_0 + a_1 x + \cdots) = 0 \end{aligned} \right\} \cdots (1)$$

Equating coefficient of x^r,

$$r(r-1) a_0 + b_0 r a_0 + c_0 a_0 = 0$$

$$\Rightarrow \quad a_0\left[r(r-1)+b_0 r+c_0\right]=0$$

$$\Rightarrow \quad \boxed{r(r-1)+b_0 r+c_0=0} \quad \text{or } r^2-(1-b_0)r+c_0=0$$

This quadratic equation is called the indicial equation and provides the r of one of the solutions and determines the form of the other solution.

So, one of the solutions of the basis is :

$$y_1 = x^{r_1}\left(a_0+a_1 x+\cdots\right)$$

The other is:

- Distinct roots (including complex roots) not differing by an integer $1, 2, \ldots$

$$y_2 = x^{r_2}\left(A_0+A_1 x+\cdots\right)$$

with coefficients obtained successively from ① with $r=r_1$ and $r=r_2$ respectively.

- A Double root

$$y_2 = y_1 \log(x) + x^{r_1}\left(A_1 x + A_2 x^2 + \cdots\right)$$

$$\left(r_2 = r_1 = \tfrac{1}{2}(1-b_0)\right)$$

- Roots differing by an integer $1, 2, \ldots$ $(r_1 > r_2)$

$$y_2 = K y_1 \log(x) + x^{r_2}\left(A_0 + A_1 x + A_2 x^2 + \cdots\right)$$

(k may turn out to be zero)

<u>Ex:</u> <u>Euler-Cauchy Equation:</u>

$$x^2y'' + b_0 x y' + c_0 y = 0 \quad (b_0, c_0 \text{ are constants})$$

Substituting $y = x^r$, we get $r(r-1)x^r + b_0 r x^r + c_0 x^r = 0$

The indicial equation is

$$r(r-1) + b_0 r + c_0 = 0 \Rightarrow r^2 + (b_0 - 1)r + c_0 = 0$$

This will give two values of r : r_1 & r_2 (say).

If $r_1 \neq r_2$, a basis of solutions : $y_1 = x^{r_1}$

$y_2 = x^{r_2}$

If $r_1 = r_2$, a basis of solutions : $y_1 = x^{r_1}$

$y_2 = x^{r_1}\log(x)$

<u>Ex:</u>

$$2x(x-1)y'' - (x+1)y' + y = 0$$

We rewrite it as $y'' - \dfrac{(x+1)}{2x(x-1)} y' + \dfrac{1}{2x(x-1)} y = 0$

$$\Rightarrow y'' + \frac{\left\{-\frac{(x+1)}{2(x-1)}\right\}}{x} y' + \frac{\frac{x}{2(x-1)}}{x^2} = 0$$

So, $b(x) = -\dfrac{(x+1)}{2(x-1)}$ and $c(x) = \dfrac{x}{2(x-1)}$

and these functions are analytic around $x=0$ and we can apply Frobenius method.

Let us substitition the Frobenius solution

$$y = x^r \sum_{m=0}^{\infty} a_m x^m \quad \text{into the ODE}$$

$$2x(x-1)y'' - (x+1)y' + y = 0$$

$$\left(\text{i.e. } 2x^2 y'' - 2x y'' - xy' - y' + y = 0\right)$$

Then $2x^2\left(x^{r-2}\sum_{m=0}^{\infty}(m+r)(m+r-1)a_m x^m\right)$

$$-2x\left(x^{r-2}\sum_{m=0}^{\infty}(m+r)(m+r-1)a_m x^m\right)$$

$$-x\left(x^{r-1}\sum_{m=0}^{\infty}(m+r)a_m x^m\right) - \left(x^{r-1}\sum_{m=0}^{\infty}(m+r)a_m x^m\right)$$

$$+\left(x^r\sum_{m=0}^{\infty}a_m x^m\right) = 0$$

$$\Rightarrow \quad 2x^r\sum_{m=0}^{\infty}(m+r)(m+r-1)a_m x^m - 2x^{r-1}\sum_{m=0}^{\infty}(m+r)(m+r-1)a_m x^m$$

$$-x^r\sum_{m=0}^{\infty}(m+r)a_m x^m - x^{r-1}\sum_{m=0}^{\infty}(m+r)a_m x^m$$

$$+x^r\sum_{m=0}^{\infty}a_m x^m = 0 \quad \ldots\ldots (1)$$

The smallest power is x^{r-1} and its coefficient gives the indicial equation

$$-2r(r-1)a_0 - r a_0 = 0$$

$$= -a_0[2r^2 - 2r + r] = 0 \Rightarrow 2r^2 - r = 0 \Rightarrow r(2r-1) = 0$$

$$\Rightarrow r = 0, \frac{1}{2}$$

Recurrence Relation: Equating the coefficient of x^{n+r}

we get $2(n+r)(n+r-1)a_n - (n+r)a_n + a_n$

$$- 2(n+1+r)(n+r)a_{n+1} - (n+1+r)a_{n+1} = 0$$

$$\Rightarrow a_{n+1}\left[2(n+1+r)(n+r) + (n+1+r)\right]$$

$$= \left[2(n+r)(n+r-1) - (n+r) + 1\right]a_n$$

$$\Rightarrow a_{n+1} = \frac{2(n+r)(n+r-1) - (n+r) + 1}{2(n+1+r)(n+r) + (n+1+r)} a_n$$

$$\Rightarrow a_{n+1} = \frac{2(n+r)(n+r-1) - (n+r) + 1}{(n+1+r)(2n+2r+1)} a_n$$

First solution:

Let $r = 0$; Then $a_{n+1} = \dfrac{2n(n-1) - n + 1}{(n+1)(2n+1)} a_n$

Now let $n=0$: Then $a_1 = \frac{1}{1} a_0 = a_0 = c_1$ (say)

let $n=1$; Then $a_2 = \frac{0-1+1}{2(3)} a_0 = 0$

Then $0 = a_3 = a_4 = \cdots$ i.e. $a_n = 0 \ \forall n \geq 2$

Hence the first solution

$$y_1 = c_1 + c_1 x = c_1(1+x)$$

<u>Second solution:</u>

Let $r = \frac{1}{2}$: Then $a_{n+1} = \frac{\left[2\left(n+\frac{1}{2}\right)\left(n-\frac{1}{2}\right) - \left(n+\frac{1}{2}\right)+1\right]}{\left(n+\frac{3}{2}\right)(2n+2)} a_n$

Let $n=0$: Then $a_1 = \frac{2\,\frac{1}{2}\times\left(-\frac{1}{2}\right) - \frac{1}{2} + 1}{\left(\frac{3}{2}\right)(2)} a_0$

$$= \frac{-1+1}{3} a_0 = 0$$

Hence $0 = a_2 = a_3 = \cdots$ i.e. $a_n = 0 \ \forall n \geq 1$

Hence the second solution is $y_2 = c_2 x^{1/2}$ $(x>0)$

Therefore the general solution $\boxed{y = c_1(1+x) + c_2\sqrt{x}}$ $(x>0)$

<u>Note:</u> The second solution can also be obtained by using reduction of order.

Ex: $(x^2-x)y'' - xy' + y = 0$.

Substituting the Frobenius Solution

$$y = x^r \sum_{m=0}^{\infty} a_m x^m \quad \text{in the ODE}$$

we get

$$(x^2-x)\left(x^{r-2}\sum_{m=0}^{\infty}(m+r)(m+r-1)a_m x^m\right)$$

$$-x\left(x^{r-1}\sum_{m=0}^{\infty}(m+r)a_m x^m\right) + \left(x^r\sum_{m=0}^{\infty}a_m x^m\right) = 0$$

$$\Rightarrow x^r\sum_{m=0}^{\infty}(m+r)(m+r-1)a_m x^m - x^{r-1}\sum_{m=0}^{\infty}(m+r)(m+r-1)a_m x^m$$

$$-x^r\sum_{m=0}^{\infty}(m+r)a_m x^m + x^r\sum_{m=0}^{\infty}a_m x^m = 0$$

$$\ldots\ldots (1)$$

The smallest power is $r-1$ whose coefficient is $\quad -r(r-1)a_0 = 0 \Rightarrow r_1 = 1,\ r_2 = 0$

We have two roots, whose difference is an integer.

First solution:

Substituting $r=1$ in (1) we obtain

$$x\sum_{m=0}^{\infty}(m+1)m\,a_m x^m - \sum_{m=0}^{\infty}(m+1)m\,a_m x^m - x.\sum_{m=0}^{\infty}(m+1)a_m x^m + x\sum_{m=0}^{\infty}a_m x^m = 0$$

$$\Rightarrow \sum_{m=0}^{\infty}(m+1)m\,a_m x^{m+1} - \sum_{m=1}^{\infty}(m+1)m\,a_m x^m - \sum_{m=0}^{\infty}(m+1)a_m x^{m+1} + \sum_{m=0}^{\infty}a_m x^{m+1} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty}(m+1)m\,a_m x^{m+1} - \sum_{m'=0}^{\infty}(m'+2)(m'+1)a_{m'+1}x^{m'+1}$$

(Let $m=m'+1$)

$$-\sum_{m=0}^{\infty}(m+1)a_m x^{m+1} + \sum_{m=0}^{\infty}a_m x^{m+1} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty}(m+1)m\,a_m x^{m+1} - \sum_{m=0}^{\infty}(m+2)(m+1)a_{m+1}x^{m+1} - \sum_{m=0}^{\infty}(m+1)a_m x^{m+1} + \sum_{m=0}^{\infty}a_m x^{m+1} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty}\Big(\big[(m+1)m-(m+1)+1\big]a_m - (m+2)(m+1)a_{m+1}\Big)x^{m+1} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty}\left[m^2 a_m - (m+2)(m+1)a_{m+1}\right]x^{m+1} = 0$$

$$\Rightarrow \quad m^2 a_m - (m+2)(m+1)a_{m+1} = 0$$

$$\Rightarrow \quad a_{m+1} = \frac{m^2}{(m+2)(m+1)} a_m$$

If we choose $a_0 = 1$, then

$$a_1 = \frac{0^2}{(0+2)(0+1)} a_0 = 0$$

Then $a_2 = a_3 = \cdots = 0$

So, $y_1 = x^{r_1} a_0 = x^1 \cdot 1 = \boxed{x}$

Second Solution:

Let us apply a reduction of order:

$$y_2 = u y_1 = u x$$

$$y_2' = u + u'x$$

$$y_2'' = u' + u' + u''x = u''x + 2u'$$

substituting in the ODE

$$(x^2 - x)(u''x + 2u') - x(xu' + u) + xu = 0$$

$$\Rightarrow (x^2-x)u''x + (x^2-x)2u' - x^2u' - \cancel{ux} + \cancel{xu} = 0$$

$$\Rightarrow (x^2-x)u'' + (x-1)2u' - xu' = 0$$

$$\Rightarrow (x^2-x)u'' + (x-2)u' = 0$$

$$\Rightarrow \frac{u''}{u'} = -\frac{(x-2)}{(x^2-x)} = -\left[\frac{2}{x} - \frac{1}{x-1}\right]$$

$$\Rightarrow \frac{u''}{u'} = -\frac{2}{x} + \frac{1}{x-1}$$

$$\Rightarrow \ln|u'| = -2\ln|x| + \ln|x-1|$$

$$= \ln\left|\frac{x-1}{x^2}\right|$$

$$\Rightarrow u' = \frac{x-1}{x^2} = \frac{1}{x} - \frac{1}{x^2}$$

$$\Rightarrow u = \ln(x) + \frac{1}{x}$$

So, $y_2 = uy_1 = \left[\ln(x) + \frac{1}{x}\right]x$

$$= \boxed{x\ln(x) + 1}$$

Hence the general solution is

$$y = c_1 x + c_2\left[x\ln(x) + 1\right]$$

Bessel Equation:

$$x^2 y'' + x y' + (x^2 - \nu^2) y = 0 \,, \quad \nu \geq 0$$

We can transform it into

$$y'' + \frac{1}{x} y' + \frac{x^2 - \nu^2}{x^2} y = 0$$

The functions 1 and $x^2 - \nu^2$ are analytic at $x = 0$ and we can use Frobenius solution

$$y(x) = x^r \sum_{m=0}^{\infty} a_m x^m = \sum_{m=0}^{\infty} a_m x^{m+r} \quad (a_0 \neq 0)$$

Substituting y'', y' and y in the Bessel Equation we get

$$\sum_{m=0}^{\infty} (m+r)(m+r-1) a_m x^{m+r} + \sum_{m=0}^{\infty} (m+r) a_m x^{m+r}$$

$$+ \sum_{m=0}^{\infty} a_m x^{m+r+2} - \nu^2 \sum_{m=0}^{\infty} a_m x^{m+r} = 0 \quad \ldots (1)$$

The smallest power is r and its coefficient gives the indicial equation

$$r(r-1) a_0 + r a_0 - \nu^2 a_0 = 0$$

$$\Rightarrow \left[r^2 - \cancel{r} + \cancel{r} - \nu^2\right] a_0 = 0 \Rightarrow (r^2 - \nu^2) a_0 = 0$$

$$\Rightarrow \quad r_1, r_2 = \pm \nu$$

Substituting $r = \nu$ in equation (1) we get

$$x^\nu \sum_{m=0}^{\infty} (m+\nu)(m+\nu-1) a_m x^m + x^\nu \sum_{m=0}^{\infty} (m+\nu) a_m x^m$$

$$+ x^{\nu+2} \sum_{m=0}^{\infty} a_m x^m - \nu^2 x^\nu \sum_{m=0}^{\infty} a_m x^m = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} \left[(m+\nu)(m+\nu-1) + (m+\nu) - \nu^2\right] a_m x^{m+\nu}$$

$$+ \sum_{m=0}^{\infty} a_m x^{m+\nu+2} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} \left[m^2 + m\nu - \cancel{m} + m\nu + \cancel{\nu^2} - \cancel{\nu} + \cancel{m} + \cancel{\nu} - \cancel{\nu^2}\right] a_m x^{m+\nu}$$

$$+ \sum_{m=0}^{\infty} a_m x^{m+\nu+2} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} m(m+2\nu) a_m x^{m+\nu} + \sum_{m=0}^{\infty} a_m x^{m+\nu+2} = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} m(m+2\nu) a_m x^{m+\nu} + \sum_{m'=2}^{\infty} a_{m'-2} x^{m'+\nu} = 0$$

$$(m' = m+2)$$

$$\Rightarrow (1+2\nu) a_1 x^{1+\nu} + \sum_{m=2}^{\infty} m(m+2\nu) a_m x^{m+\nu} + \sum_{m=2}^{\infty} a_{m-2} x^{m+\nu} = 0$$

$$\Rightarrow (1+2\nu) a_1 x^{1+\nu} + \sum_{m=0}^{\infty} \left[m(m+2\nu) a_m + a_{m-2} \right] x^{m+\nu} = 0$$

$$\Rightarrow (1+2\nu) a_1 = 0 \quad \Rightarrow a_1 = 0$$

and $\quad m(m+2\nu) a_m + a_{m-2} = 0$

$$\Rightarrow a_m = -\frac{1}{m(2\nu+m)} a_{m-2}$$

Now if m is odd, $\quad 0 = a_1 = a_3 = a_5 = \dots$

If m is even, we can write $m = 2k$

and $\quad a_{2k} = -\dfrac{1}{2k(2\nu+2k)} a_{2k-2} = -\dfrac{1}{2^2 k(\nu+k)} a_{2k-2}$

for $k = 1, 2, \dots$

Thus $\quad a_2 = -\dfrac{1}{2^2(\nu+1)} a_0$

$$a_4 = -\frac{1}{2^2\, 2\,(\nu+2)} a_2 = -\frac{1}{2^2\, 2\,(\nu+2)} \left(-\frac{1}{2^2(\nu+1)} a_0 \right)$$

$$= \frac{1}{2^4\, 2!\, (\nu+1)(\nu+2)} a_0$$

In general, $\quad a_{2k} = \dfrac{(-1)^m}{2^{2k} k!\, (\nu+1)(\nu+2)\cdots(\nu+k)} a_0$

The first solution of Bessel equation is

$$y_1 = x^{\nu} \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{2k} k!\, (\nu+1)(\nu+2)\cdots(\nu+k)} a_0 x^{2k}$$

If ν is an integer let $\nu = n$

Let us choose $a_0 = \frac{1}{2^n n!}$

Then $$y_1 = x^n \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{2k} k! (n+1)(n+2) \cdots (n+k)} \frac{1}{2^n n!} x^{2k}$$

$$\Rightarrow y_1 = \boxed{J_n = x^n \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{2k+n} k! (n+k)!} x^{2k}}$$

- This is the <u>Bessel's function of the first kind and order n</u> (The series converges for all x by ratio test)

- For large x, they fullfill

$$J_n(x) \sim \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{n\pi}{2} - \frac{\pi}{4}\right)$$

(asymptotically equal)

<u>The Gamma Function</u>:

Define gamma function as:

$$\Gamma(x+1) = \int_0^{\infty} e^{-t} t^x \, dt \qquad (\text{for } x > -1)$$

Integrating by parts we get

$$\Gamma(x+1) = -e^{-t} t^x \Big|_0^{\infty} + \int_0^{\infty} x t^{x-1} e^{-t} \, dt$$

$$\Rightarrow \Gamma(x+1) = 0 + x\int_0^\infty e^{-t} t^{x-1}\, dt = x\Gamma(x)$$

$$\Rightarrow \boxed{\Gamma(x+1) = x\Gamma(x)}$$

Now substituting $x=0$ in the definition

$$\Gamma(1) = \int_0^\infty e^{-t} dt = 1 \qquad \Rightarrow \boxed{\Gamma(1) = 1}$$

Thus for any nonnegative integer n

$$\boxed{\Gamma(n+1) = n!} \qquad n = 0, 1, 2, \ldots$$

Hence the gamma function generalizes the factorial function to arbitrary positive ν.

Another important result is $\boxed{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}$

Bessel function of first kind $J_\nu(x)$:

The first solution of Bessel equation is:

$$y_1 = x^\nu \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{2k} k!\,(\nu+1)(\nu+2)\ldots(\nu+k)} a_0 x^{2k}$$

If ν is not an integer, let $a_0 = \dfrac{1}{2^\nu \Gamma(\nu+1)}$

Then $$y_1 = x^{\nu} \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{2k} k! (\nu+1)(\nu+2)\cdots(\nu+k)} \frac{1}{2^{\nu}\Gamma(\nu+1)} x^{2k}$$

$$\Rightarrow \quad y_1 = \boxed{J_\nu(x) = x^{\nu} \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{2k+\nu} k! \, \Gamma(\nu+k+1)} x^{2k}}$$

- $J_\nu(x)$ is called the Bessel function of the first kind of order ν. The series converges for all x by ratio test.

- An interesting result is : $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin(x)$

  and $J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos(x)$

- If ν is not an integer, then the general solution is : $\boxed{y = c_1 J_\nu + c_2 J_{-\nu}}$

  where $$J_{-\nu} = x^{-\nu} \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{2k-\nu} k! \, \Gamma(-\nu+k+1)} x^{2k}$$

$\nu = n$

- If ν is an integer, there is a problem because $J_{-n} = (-1)^n J_n$

that is, there is a linear dependence between the two solutions. (This will be resolved by

Bessel function of second kind)

- **Some Useful Properties:**

Derivatives: (1) $\left[x^{\nu} J_\nu(x)\right]' = x^{\nu} J_{\nu-1}(x)$

(2) $\left[x^{-\nu} J_\nu(x)\right]' = -x^{-\nu} J_{\nu+1}(x)$

Recursion: (3) $J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x)$

(4) $J_{\nu-1}(x) - J_{\nu+1}(x) = 2 J_\nu'(x)$

Bessel Equation:

Let us find a second solution of Bessel equation in case ν is an integer.

First we assume $\nu = n = 0$

Then the ODE becomes: $x^2 y'' + xy' + x^2 y = 0$

$$\Rightarrow \quad xy'' + y' + xy = 0$$

Then the indicial equation becomes $r^2 = 0$

i.e. it has a double root at $r = 0$.

The first solution is $\boxed{y_1 = J_0}$

The second solution according to Frobenius method must be of the form:

$$y_2 = y_1 \ln(x) + x^r \sum_{m=1}^{\infty} A_m x^m$$

$$\Rightarrow y_2 = J_0 \ln(x) + \sum_{m=1}^{\infty} A_m x^m \qquad (r=0)$$

$$\Rightarrow y_2' = J_0' \ln(x) + J_0 \frac{1}{x} + \sum_{m=1}^{\infty} m A_m x^{m-1}$$

$$y_2'' = J_0'' \ln(x) + 2 J_0' \frac{1}{x} - J_0 \frac{1}{x^2} + \sum_{m=1}^{\infty} m(m-1) A_m x^{m-2}$$

Substituting in the ODE $xy'' + y' + xy = 0$

we get

$$x\left(J_0'' \ln(x) + 2J_0' \frac{1}{x} - J_0 \frac{1}{x^2} + \sum_{m=1}^{\infty} m(m-1) A_m x^{m-2}\right)$$

$$+\left(J_0' \ln(x) + J_0 \frac{1}{x} + \sum_{m=1}^{\infty} m A_m x^{m-1}\right)$$

$$+ x\left(J_0 \ln(x) + \sum_{m=1}^{\infty} A_m x^m\right) = 0$$

$$\Rightarrow \left(xJ_0'' + J_0' + xJ_0\right) \ln(x) + 2J_0' + \sum_{m=1}^{\infty} m(m-1) A_m x^{m-1}$$

$$+ \sum_{m=1}^{\infty} m A_m x^{m-1} + \sum_{m=1}^{\infty} A_m x^{m+1} = 0$$

$$\Rightarrow \quad 2J_0' + \sum_{m=1}^{\infty} m^2 A_m x^{m-1} + \sum_{m=1}^{\infty} A_m x^{m+1} = 0 \quad \ldots\ldots (1)$$

Now $\quad J_n = x^n \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{2k+n} \, k! \, (n+k)!} x^{2k}$

In particular $\quad J_0 = \sum_{m=0}^{\infty} \frac{(-1)^m}{2^{2m} (m!)^2} x^{2m}$

$$J_0' = \sum_{m=1}^{\infty} \frac{(-1)^m \, 2m}{2^{2m} (m!)^2} x^{2m-1} = \sum_{m=1}^{\infty} \frac{(-1)^m}{2^{2m-1} m! \, (m-1)!} x^{2m-1}$$

Substituting in (1) we get

$$2\left(\sum_{m=1}^{\infty} \frac{(-1)^m}{2^{2m-1} m! \, (m-1)!} x^{2m-1} \right) + \sum_{m=1}^{\infty} m^2 A_m x^{m-1} + \sum_{m=1}^{\infty} A_m x^{m+1}$$

$$= 0$$

$$\Rightarrow \sum_{m=1}^{\infty} \frac{(-1)^m}{2^{2m-2} m! \, (m-1)!} x^{2m-1} + \sum_{m=1}^{\infty} m^2 A_m x^{m-1}$$

$$+ \sum_{m=1}^{\infty} A_m x^{m+1} = 0$$

$$\Rightarrow \left(-x + \frac{1}{2^2 \, 2!} x^3 - \ldots \right) + \left(A_1 + 4A_2 x + \ldots\ldots \right)$$

$$+ \left(A_1 x^2 + A_2 x^3 + \ldots\ldots \right) = 0$$

Now equating powers of x^0 from both sides we get $A_1 = 0$

Now consider even powers x^{2s}

First series has no even power.

In the second series $m-1 = 2s \Rightarrow m = (2s+1)$

and the term is $(2s+1)^2 A_{2s+1} x^{2s}$

In the third series $m+1 = 2s \Rightarrow m = 2s-1$

and the term is $A_{2s-1} x^{2s}$

Hence $(2s+1)^2 A_{2s+1} + A_{2s-1} = 0$ for $s = 1, 2, \ldots$

$$\Rightarrow A_{2s+1} = -\frac{1}{(2s+1)^2} A_{2s-1}$$

Since $A_1 = 0$, we get $0 = A_3 = A_5 = \cdots\cdots$

Now consider the coefficients of x^{2s+1}

For $s = 0$, we get $-1 + 4A_2 = 0 \Rightarrow A_2 = \frac{1}{4}$

For other values of s, in the first series $2s+1 = 2m-1$
$\Rightarrow m = s+1$

in the second series $2s+1 = m-1 \Rightarrow m = 2s+2$

in the third series $2s+1 = m+1 \Rightarrow m = 2$

Therefore $\dfrac{(-1)^{s+1}}{2^{2s}(s+1)!\,s!} + (2s+2)^2 A_{2s+2} + A_{2s} = 0$

For $s=1$, we get $\frac{1}{8} + 16A_4 + A_2 = 0$ $\left(A_2 = \frac{1}{4}\right)$

$$\Rightarrow A_4 = -\frac{3}{128}$$

In general $A_{2m} = \dfrac{(-1)^{m-1}}{2^{2m}(m!)^2}\left(1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{m}\right)$

for $m = 1, 2, \ldots$

Let us use the notation: $h_1 = 1$, $h_m = 1 + \frac{1}{2} + \cdots + \frac{1}{m}$

for $m = 2, 3, \ldots$

Then $y_2(x) = J_0(x)\ln(x) + \displaystyle\sum_{m=1}^{\infty} \frac{(-1)^{m-1} h_m}{2^{2m}(m!)^2} x^{2m}$

$$= J_0(x)\ln(x) + \frac{1}{4}x^2 - \frac{3}{128}x^4 + \frac{11}{13824}x^6 - \cdots$$

Since J_0 and y_2 are linearly independent functions they form a basis of solutions for $x > 0$.

It is customary to use a different basis

where y_2 will be replaced by an independent particular solution $Y_0 = a(y_2 + bJ_0)$ where $a (\neq 0)$

and b are constants chosen as $a_0 = \frac{2}{\pi}$

and $b = \gamma - \ln 2$

where γ is Euler's constant

$$\gamma = \lim_{s \to \infty} \left(1 + \frac{1}{2} + \cdots + \frac{1}{s} - \ln s\right) \approx .5772$$

Thus $Y_0(x) = \frac{2}{\pi}\left[y_2 + (\gamma - \ln(2)) J_0(x)\right]$

$$= \frac{2}{\pi}\left[J_0(x)\left(\ln\left(\frac{x}{2}\right) + \gamma\right) + \sum_{m=1}^{\infty} \frac{(-1)^{m-1} h_m}{2^{2m}(m!)^2} x^{2m}\right]$$

and $J_0(x)$ and $Y_0(x)$ forms a basis of solutions.

- $Y_0(x)$ is called the Bessel function of the second kind of order zero. (or Neumann's function of order zero)

- For small $x > 0$, $Y_0(x)$ behaves like $\log(x)$ and $Y_0(x) \longrightarrow -\infty$ as $x \to 0$

- In general,

$$Y_\nu(x) = \frac{1}{\sin(\nu\pi)}\left[J_\nu(x)\cos\nu\pi - J_{-\nu}(x)\right]$$

and $Y_n(x) = \lim_{\nu \to n} Y_\nu(x)$

This function is called the Bessel function of second kind of order ν (or Neumann's function of order ν).

- J_ν and Y_ν are linearly independent for all ν (and $x > 0$)

- Therefore a general solution of Bessel Equation

$$x^2 y'' + x y' + (x^2 - \nu^2) y = 0 \quad (\nu \geq 0)$$

can be given by $\boxed{y(x) = c_1 J_\nu(x) + c_2 Y_\nu(x)}$

Note: Sometimes we need solutions of Bessel Equation that are complex for real values of x. In that case the solutions:

$$\left.\begin{aligned} H_\nu^{(1)}(x) &= J_\nu(x) + i Y_\nu(x) \\ H_\nu^{(2)}(x) &= J_\nu(x) - i Y_\nu(x) \end{aligned}\right\}$$

are used.

These linearly independent functions are called Bessel functions of third kind of order ν or first and second Hankel functions of order ν

- For large x, they satisfy

$$Y_n(x) \sim \sqrt{\frac{2}{\pi x}} \sin\left(x - \frac{n\pi}{2} - \frac{\pi}{4}\right)$$

- There are related recurrence formulas and formulas for derivatives of $J_\nu(x)$ and $Y_\nu(x)$

Modified Bessel Equation:

Bessel Equation: $x^2y'' + xy' + (x^2-\nu^2)y = 0$

Modified Bessel Equation: $x^2y'' + xy' - (x^2+\nu^2)y = 0$

- The general solution of the Modified Bessel Equation is of the form:

$$y = c_1 I_\nu + c_2 K_\nu$$

where I_ν is the modified Bessel function of first kind

$$I_\nu(x) = i^{-\nu} J_\nu(ix) = x^\nu \sum_{k=0}^{\infty} \frac{1}{2^{2k+\nu} k! \, \Gamma(k+\nu+1)} x^{2k}$$

(Compare it to: $J_\nu(x) = x^\nu \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{2k+\nu} k! \, \Gamma(\nu+k+1)} x^{2k}$)

and K_ν is the modified Bessel function of second kind:

$$K_\nu(x) = \frac{\pi}{2} \cdot \frac{I_{-\nu}(x) - I_\nu(x)}{\sin \nu\pi}$$

(Compare it to $Y_\nu(x) = \frac{J_\nu(x)\cos(\nu\pi) - J_{-\nu}(x)}{\sin \nu\pi}$)